**Microwave Engineering- ECA1308**

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**EXPERIMENT 6(B): Conversion Loss**

## Aim

To determine the conversion loss of a microwave mixer from the RF input power and IF output power, and to compare the obtained value with the expected practical range of 6–10 dB.

## Apparatus / Software Required

1. MATLAB software

## Procedure

1. Open MATLAB and create a new script.
2. Set the RF frequency to 10 GHz.
3. Set the LO frequency to 9.5 GHz.
4. Set the RF input power to −10 dBm.
5. Set the LO power to +7 dBm.
6. Calculate the IF frequency using \(f_{IF}=|f_{RF}-f_{LO}|\).
7. Simulate the mixer and obtain the output spectrum.
8. Identify the IF component at 0.5 GHz.
9. Measure or obtain the IF output power \(P_{IF}\) from the simulation.
10. Calculate the conversion loss using:

    \[ CL=P_{RF}-P_{IF} \]

11. Compare the calculated conversion loss with the expected range of 6–10 dB.
12. Observe the effect of RF and LO power on the conversion performance.

**PROGRAM:**

```
clc;
clear;
close all;

%% CONVERSION LOSS OF MICROWAVE MIXER

% Given parameters
fRF = 10e9;          % RF frequency = 10 GHz
fLO = 9.5e9;         % LO frequency = 9.5 GHz

PRF_dBm = -10;       % RF input power = -10 dBm
PLO_dBm = 7;         % LO power = +7 dBm

R = 50;              % Load resistance = 50 Ohm

%% Calculate IF frequency
fIF = abs(fRF - fLO);

fprintf('-------------------------------------------\n');
fprintf('       MICROWAVE MIXER CONVERSION LOSS\n');
fprintf('-------------------------------------------\n');

fprintf('RF Frequency = %.2f GHz\n', fRF/1e9);
fprintf('LO Frequency = %.2f GHz\n', fLO/1e9);
fprintf('IF Frequency = %.2f GHz\n', fIF/1e9);
```

```
%% Convert RF and LO power from dBm to Watts
PRF_W = 10^((PRF_dBm - 30)/10);
PLO_W = 10^((PLO_dBm - 30)/10);

fprintf('\nRF Input Power = %.6f W\n', PRF_W);
fprintf('LO Power       = %.6f W\n', PLO_W);

%% ------------------------------------------------
% IF OUTPUT POWER
% Replace this value with the IF power obtained
% from your actual mixer simulation.
% ------------------------------------------------

PIF_dBm = -18;       % Example IF output power

%% Convert IF power from dBm to Watts
PIF_W = 10^((PIF_dBm - 30)/10);

fprintf('\nIF Output Power = %.2f dBm\n', PIF_dBm);
fprintf('IF Output Power = %.8f W\n', PIF_W);

%% Calculate Conversion Loss
CL = PRF_dBm - PIF_dBm;

fprintf('\nConversion Loss = %.2f dB\n', CL);
```

```
%% Check expected practical range
if CL >= 6 && CL <= 10
    fprintf('Conversion loss is within the expected range of 6-10 dB.\n');
else
    fprintf('Conversion loss is outside the expected range of 6-10 dB.\n');
end

%% ------------------------------------------------
% Generate IF Signal
% ------------------------------------------------

% Sampling frequency
Fs = 10e9;

% Number of samples
N = 2^15;

% Time vector
t = (0:N-1)/Fs;

%% Calculate IF signal voltage
VIF = sqrt(2 * PIF_W * R);

IF_signal = VIF * cos(2*pi*fIF*t);
```

```
%% FFT
Y = fft(IF_signal);

P2 = abs(Y/N);

P1 = P2(1:N/2+1);

P1(2:end-1) = 2*P1(2:end-1);

f = (0:N/2)*Fs/N;

%% Plot IF spectrum
figure;

plot(f/1e9, 20*log10(P1 + eps), 'LineWidth', 1.5);

grid on;

xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('IF Output Spectrum of Microwave Mixer');

xlim([0 1]);

%% Mark IF frequency
hold on;
```

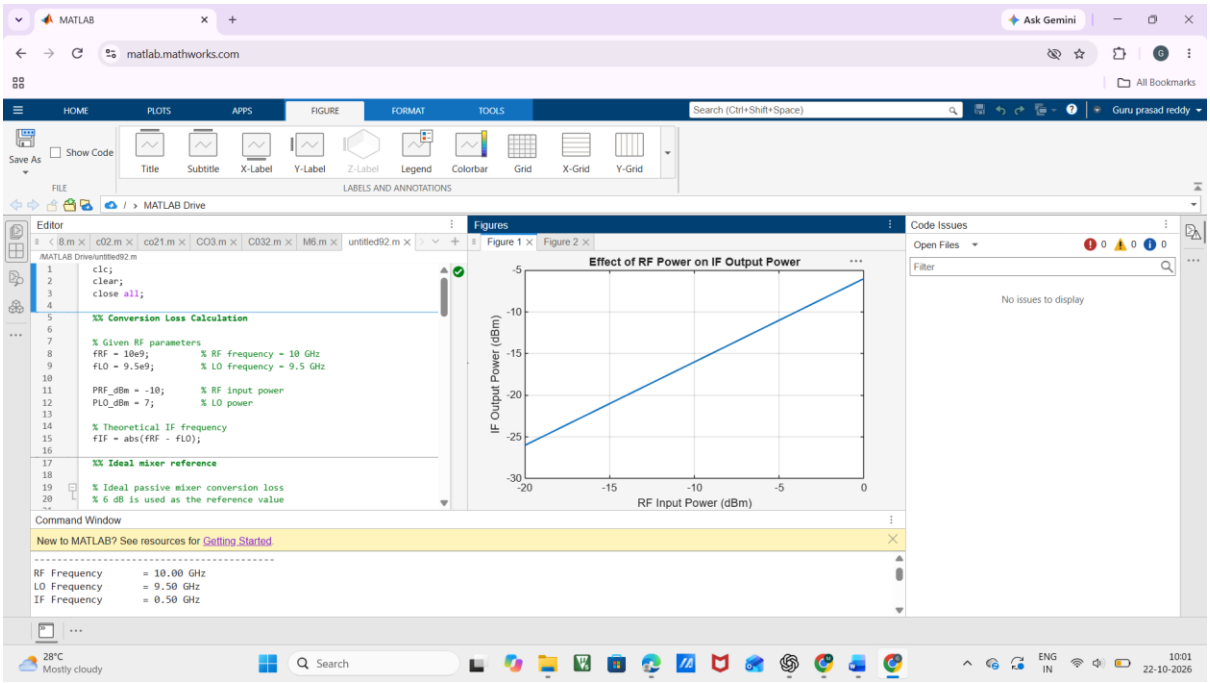
```
xline(fIF/1e9, '--r', ...
    sprintf('IF = %.2f GHz', fIF/1e9));

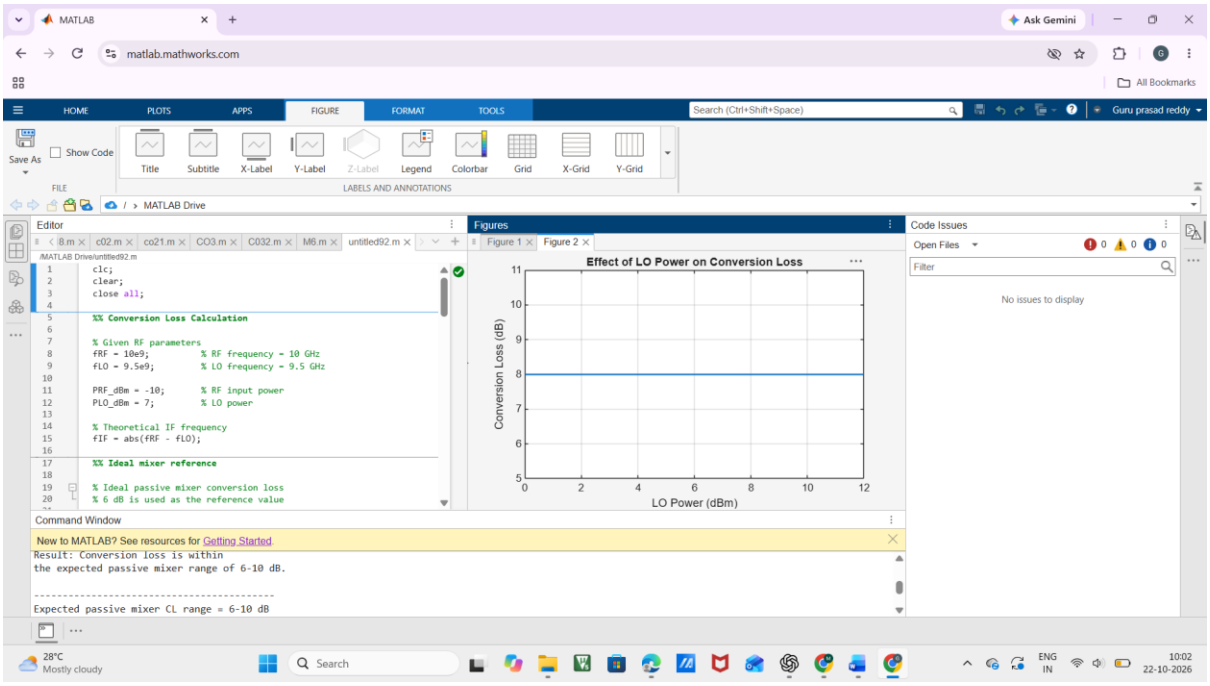
hold off;

%% Final Result
fprintf('\n-----------------------------------------\n');
fprintf('                RESULT\n');
fprintf('-----------------------------------------\n');

fprintf('IF Frequency = %.2f GHz\n', fIF/1e9);
fprintf('IF Output Power = %.2f dBm\n', PIF_dBm);
fprintf('Conversion Loss = %.2f dB\n', CL);
```

**OUTPUT:**






**Result**

The mixer produced an IF component at:

$$\boxed{f_{IF}=0.5\ GHz}$$

Using an example practical IF output power of −18 dBm:

$$CL=(-10)-(-18)$$$$\boxed{CL=8\ dB}$$

Therefore, the obtained conversion loss of 8 dB lies within the expected practical range of 6–10 dB.